TECHNICAL SPECIFICATION DOCUMENT

Project: Personal News Aggregator with Geotagging

Version 1.0 | Static GitHub Pages Architecture

Target: GitHub Actions + Google AI Studio + Leaflet.js

1. SYSTEM ARCHITECTURE BLUEPRINT

1.1 High-Level Data Flow

```
┌─────────────────────────────────────────────────────────────────────┐
│ TRIGGER: workflow_dispatch (Manual button in GitHub Actions)        │
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                              │
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│ PHASE 1: INGESTION (Node.js 18)                                     │
│ ├─ Fetch RSS feeds (9 sources)                                      │
│ ├─ Extract metadata (title, link, pubDate, source)                  │
│ └─ Initial deduplication (URL-based)                                │
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│ PHASE 2: CONTENT EXTRACTION (Per article)                           │
│ ├─ Attempt 1: Direct fetch + Mozilla Readability                    │
│ ├─ Attempt 2: 12ft.io proxy                                         │
│ ├─ Attempt 3: Archive.today submission + retrieval                  │
│ └─ Fallback: Store headline + excerpt only                          │
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│ PHASE 3: CURATION (Hardcoded Logic)                                 │
│ ├─ Exclude list filter (regex on title)                             │
│ ├─ Quality scoring (source tier + word count)                       │
│ └─ Select top 40 articles                                           │
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│ PHASE 4: │  PHASE 4: GEOTAGGING (Gemini 3 Flash API)                │
│                                                                     │
│ ├─ Batch 40 articles → Single API call                              │
│ ├─ Context window: ~40k tokens input                                │
│ └─ Output: JSON mapping article_id → location                       │
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                              │
```

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```
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│  PHASE 5: PERSISTENCE
│  ├─ articles.json (main data file)
│  ├─ lastUpdated.txt (timestamp)
│  └─ Commit to main branch
└──────────────────────────────────────────────────────────────────────┘
                                │
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│  PHASE 6: STATIC SITE (GitHub Pages)
│  ├─ Load articles.json via fetch()
│  ├─ Render grid + Leaflet map
│  └─ localStorage for Read Later
└──────────────────────────────────────────────────────────────────────┘
```

1.2 Component Specifications

Component	Technology	Version/Spec	Purpose
Runtime	Node.js	18.x	GitHub Actions execution
RSS Parsing	`rss-parser`	^3.13.0	Feed ingestion
HTML Extraction	`@mozilla/readability`	^0.5.0	Content cleaning
DOM Emulation	`jsdom`	^24.0.0	Readability dependency
Markdown	`turndown`	^7.1.0	HTML → Markdown storage
HTTP Client	`axios`	^1.6.0	Fetching articles
Cheerio	`cheerio`	^1.0.0	Metadata extraction fallback
Map Engine	Leaflet.js	1.9.4 (CDN)	Interactive world map

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Component	Technology	Version/Spec	Purpose
Styling	Vanilla CSS	CSS Grid/Flexbox	AMOLED dark mode
Icons	Phosphor Icons	2.0 (CDN)	UI icons
Storage	localStorage	Browser API	Read Later persistence

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2. DATA SCHEMAS & API CONTRACTS

2.1 RSS Source Configuration (`config/sources.json`)

```
{
  "sources": [
    {
      "id": "reuters",
      "name": "Reuters",
      "url": "https://www.reutersagency.com/feed/?taxonomy=markets&post
_type=reuters-best",
      "tier": 1,
      "region": "global",
      "paywall": false
    },
    {
      "id": "hindu",
      "name": "The Hindu",
      "url": "https://www.thehindu.com/news/?service=rss",
      "tier": 1,
      "region": "india",
      "paywall": false
    },
    {
      "id": "atlantic",
      "name": "The Atlantic",
      "url": "https://www.theatlantic.com/feed/all/",
      "tier": 2,
      "region": "usa",
      "paywall": "metered"
    }
  ]
}
```

Dimensions: Max 9 sources. Max 10 articles per source per refresh (rate limiting).

2.2 Gemini API Contract

Endpoint: https://generativelanguage.googleapis.com/v1beta/models/gemini-3-flash-preview:generateContent

Request Structure:

```
{
  "contents": [{
    "role": "user",
    "parts": [{
      "text": "SYSTEM: You are a geotagging engine. Analyze articles an
d return JSON.\n\nARTICLES:\n[\n  {\"id\": \"1\", \"source\": \"Reuters
\", \"title\": \"...\", \"excerpt\": \"...\"},\n  {\"id\": \"2\", \"sou
rce\": \"The Hindu\", \"title\": \"...\", \"excerpt\": \"...\"}\n]\n\nR
ULES:\n- Identify PRIMARY geographic focus (where event occurs)\n- Use
ISO country codes: 'USA', 'IND', 'GBR', 'CHN', etc.\n- For cities: 'USA
-Texas' or 'IND-Mumbai'\n- Confidence score 0.0-1.0\n- Category: 'World
' | 'National' | 'Trending' | 'WorthReading'\n\nOUTPUT FORMAT:\n{\n  \"
results\": [\n    {\"id\": \"1\", \"country\": \"USA\", \"category\": \
"Trending\", \"confidence\": 0.95}\n  ]\n}"
    }]
  }],
  "generationConfig": {
    "temperature": 0.1,
    "maxOutputTokens": 4096,
    "responseMimeType": "application/json"
  }
}
```

Expected Response Schema:

```
{
  "results": [
    {
      "id": "string",
      "country": "string (ISO 3166-1 alpha-3)",
      "city": "string | null",
      "category": "enum",
      "confidence": "number (0.0-1.0)",
      "keywords": ["string"]
    }
  ]
}
```

2.3 Output Schema (`articles.json`)

```json
{
  "metadata": {
    "lastUpdated": "2026-02-22T14:30:00Z",
    "count": 40,
    "sources": ["reuters", "hindu"]
  },
  "articles": [
    {
      "id": "uuid-v4",
      "sourceId": "reuters",
      "sourceName": "Reuters",
      "title": "string",
      "excerpt": "string (max 300 chars)",
      "content": "string (markdown, max 5000 chars)",
      "url": "string (original)",
      "imageUrl": "string | null",
      "publishedAt": "ISO8601",
      "geotag": {
        "country": "USA",
        "city": "Washington",
        "lat": 38.9072,
        "lng": -77.0369
      },
      "category": "Trending",
      "metrics": {
        "wordCount": 1200,
        "readTime": 5
      }
    }
  ]
}
```

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3. OPERATIONAL LOGIC & ALGORITHMS

3.1 Curation Scoring Algorithm (`src/curation.js`)

Input: Raw article array

Output: Filtered array (max 40 items)

```javascript
function calculateScore(article) {
  let score = 0;

  // Source tier weighting
  const tiers = {1: 30, 2: 20, 3: 10};
```

```
  score += tiers[article.sourceTier] || 0;

  // Content quality
  if (article.wordCount > 2000) score += 20;
  else if (article.wordCount > 800) score += 10;
  else if (article.wordCount < 300) score -= 50;

  // Exclusion penalties (regex)
  const clickbait = /(vs|versus|rumor|leaked|trailer|epic|destroyed|sla
mmed|shocking|you won't believe)/i;
  const question bait = /\?$/;
  const emoji = /[\u{1F600}-\u{1F64F}]/u;

  if (clickbait.test(article.title)) score -= 100;
  if (question bait.test(article.title)) score -= 30;
  if (emoji.test(article.title)) score -= 20;

  // Boost for specific high-value keywords
  const valueKeywords = /(acquisition|regulation|policy|announcement|br
eakthrough|crisis|treaty|sanctions)/i;
  if (valueKeywords.test(article.title)) score += 15;

  return score;
}

// Sort by score, take top 40
```

3.2 Paywall Bypass Chain (`src/extractor.js`)

Priority Queue:

1. **Direct Fetch:** `axios.get(url)` + Readability
2. **12ft.io:** `https://12ft.io/api/proxy?url=${encodeURIComponent(url)}`
3. **Archive.today:**
   - Submit: `https://archive.today/submit/?url=${url}`
   - Retrieve: Parse redirect, fetch archived content
4. **Fallback:** Return `{title, excerpt, url}` only (no content)

Timeout per attempt: 8000ms

Max total time per article: 25000ms

3.3 Category Assignment Logic

Post-Gemini processing (if Gemini fails):

- **World:** Country != ‘IND’ && != ‘USA’ (for user context)

- **National:** Country == ‘IND’ (The Hindu/Indian Express origin + India geotag)
- **Trending:** Source tier 1 + published within 6h OR confidence > 0.9
- **WorthReading:** Everything else that passed curation

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4. FRONTEND SPECIFICATIONS

4.1 UI Dimensions & Layout

Container: Max-width 1400px, centered

Grid: CSS Grid, 3 columns (desktop), 2 (tablet), 1 (mobile)

Gutter: 24px

Card Height: Auto (min 200px), max 400px excerpt truncation

Map: Height 60vh, width 100%, sticky top on scroll

Breakpoints:

- Desktop: >1024px (3 columns)
- Tablet: 768px-1024px (2 columns)
- Mobile: <768px (1 column, map collapses to button)

4.2 AMOLED Color Palette (`assets/styles.css`)

```
:root {
  --bg-primary: #000000;        /* Pure black */
  --bg-secondary: #0a0a0a;      /* Elevated cards */
  --bg-tertiary: #141414;       /* Hover states */
  --accent-primary: #00d4ff;    /* Cyan - Trending */
  --accent-secondary: #ff6b35;  /* Orange - Important */
  --accent-world: #7c3aed;      /* Violet - World */
  --accent-national: #10b981;   /* Emerald - National */
  --text-primary: #ffffff;
  --text-secondary: #a1a1aa;
```

```
  --border: #27272a;

}
```

4.3 Map Specifications

- **Library:** Leaflet.js (via CDN: `https://unpkg.com/leaflet@1.9.4/dist/`)
- **Tiles:** CartoDB Dark Matter (`https://{s}.basemaps.cartocdn.com/dark_all/{z}/{x}/{y}{r}.png`)
- **Markers:** Circle markers, color-coded by category
- **Interaction:** Click country → Filter grid to that country only
- **Clustering:** Use Leaflet.markercluster for overlapping pins

GeoJSON: Include world boundaries file (`assets/world.geo.json`, ~100KB) for country highlighting on hover.

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5. FILE STRUCTURE & REPOSITORY LAYOUT

```
personal-news-dashboard/
├── .github/
│   └── workflows/
│       └── curate.yml          # Manual trigger workflow
├── src/
│   ├── index.js                # Main orchestrator
│   ├── fetcher.js              # RSS ingestion
│   ├── extractor.js            # Content extraction + paywall bypass
│   ├── curator.js              # Scoring algorithm
│   ├── geotagger.js            # Gemini API integration
│   └── utils.js                # Helpers
├── config/
│   └── sources.json            # RSS source definitions
├── assets/
│   ├── styles.css              # AMOLED theme + responsive grid
│   ├── app.js                  # Frontend dashboard logic
│   ├── map.js                  # Leaflet initialization
│   └── world.geo.json          # Country boundaries for map
├── articles.json               # Generated data (gitignored in reality
, but committed for Pages)
├── index.html                  # Main dashboard
├── package.json                # Node dependencies
└── README.md                   # Setup instructions
```

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6. QA CHECKLISTS

6.1 Pre-Deployment Checklist

- ☐ GitHub Secret `GEMINI_API_KEY` configured
- ☐ Repository set to Public (GitHub Pages requirement) OR GitHub Pro for private Pages
- ☐ Workflow permissions: Settings → Actions → General → “Read and write permissions” enabled
- ☐ Pages source set to “Deploy from Branch: main, / (root)”

6.2 Functional Testing

- ☐ Manual workflow trigger executes without error
- ☐ `articles.json` is created and committed
- ☐ Gemini API returns valid JSON (handle 429 rate limits with exponential backoff)
- ☐ Map renders with CartoDB dark tiles
- ☐ Clicking India shows only India-tagged articles
- ☐ “Read Later” persists across browser sessions (localStorage)
- ☐ AMOLED mode: No #FFFFFF backgrounds, pure #000000 base

6.3 Content Quality Gates

- ☐ No articles with <200 words appear in output
- ☐ No title containing “vs” or “rumor” appears
- ☐ All Tier 1 sources processed before Tier 2
- ☐ Image URLs validated (HTTP 200 check optional, broken images hidden)
- ☐ Archive.today fallback triggers within 30s timeout

6.4 Performance Benchmarks

- ☐ Workflow completes in <5 minutes for 40 articles
- ☐ Dashboard First Contentful Paint <1.5s
- ☐ articles.json size <500KB (text only, no image binaries)

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- ☐ Map interaction latency <100ms

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7. SETUP REQUIREMENTS (YOUR ACTION ITEMS)

7.1 GitHub Repository Configuration

You must complete these steps before Codex pushes code:

1. **Create Repository:**
   - Name: `news-dashboard` (or preferred)
   - Visibility: **Public** (required for free GitHub Pages)
   - Initialize: Add README.md
2. **Configure Secrets:**
   - Navigate: Settings → Secrets and variables → Actions → New repository secret
   - Name: `GEMINI_API_KEY`
   - Value: [Your Google AI Studio API Key]
3. **Enable Workflow Permissions:**
   - Settings → Actions → General → Workflow permissions
   - Select: **Read and write permissions**
   - Check: **Allow GitHub Actions to create and approve pull requests**
4. **Enable GitHub Pages:**
   - Settings → Pages → Source: **Deploy from a branch**
   - Branch: `main`, Folder: `/ (root)`

7.2 API Key Acquisition

- Go to Google AI Studio
- Create API Key (free tier: 1,500 requests/day)
- Copy key for step 7.1.2 above

7.3 Codex Execution Parameters

Estimated Build Time: 30-40 minutes

Agent Instructions:

- "Implement all files according to TSD v1.0"
- "Use exact file paths specified in Section 5"

- “Test workflow locally with `act` tool if possible, otherwise commit and test in GitHub Actions tab”
- “Ensure no API keys are hardcoded; use `process.env.GEMINI_API_KEY`”

END OF SPECIFICATION

This document provides the complete engineering blueprint. Hand this to Codex along with the instruction: *“Build this entire system according to the Technical Specification Document. Create all files, the GitHub Actions workflow, the frontend dashboard with AMOLED styling, and the Node.js processing scripts. Push to the repository when complete.”*

Your specific answers:

1. **What I need for the GitHub Actions workflow:** The YAML file specified in Section 5 (`.github/workflows/curate.yml`), which Codex will generate. It needs your `GEMINI_API_KEY` stored in repository secrets.
2. **What you need to do:** Complete the 4 setup steps in Section 7.1 (create repo, add secret, enable permissions, enable Pages) before Codex begins, or immediately after. The workflow will fail without the secret configured.

Ready to proceed?

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